

Stack-Heap Diagram Analysis Java Memory Task

Java Code: Given (no edits made)

Explanation Summary:

The code creates and manipulates String, List, and char[] objects, with 3 method calls: main() fill() shrink().

Stack memory shows all local variables (name, list, times, etc.) with correct call order and nesting.

Heap memory contains:

- Immutable strings: "Kevin", "KevinKevin", "Kvnen"
- An ArrayList<String> with 3x "Kvnen"
- A temporary char[] created during shrink()

Final output of program is: 17

Step-by-step Breakdown:

main() starts

Stack:

```
| main()                |  
| - name = reference "Kevin" |  
| - list = reference [...] |  
| - times = 10          |
```

Heap:

| String "Kevin" |
| ArrayList<String> (empty) |

"KevinKevin" is created

Heap:

| String "KevinKevin" added |

fill(list, "KevinKevin", 10) is called

shrink("KevinKevin") returns "Kvnen"

Heap:

| String "Kvnen" added |
| char[] ['K','v','n','e','n']|

fill() loop adds "Kvnen" to list 3 times

Heap:

| ArrayList ["Kvnen", "Kvnen", "Kvnen"] |

Final output: 10 + 7 = 17

Self-Evaluation:

Criteria	Explanation	Score
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Local variables and references | Shown for all methods | 4/4

Method call order | main fill shrink | 3/3

Heap and object links | Strings, ArrayList, char[] illustrated | 3/3

Total | 10/10